HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

FINAL PROJECT REPORT

Topic: Parallelization Methods for CNN-Based Computer Vision Inference

Course: Parallel and Distributed Programming

Course code: IT4130E

Class/session: 168069 - Friday morning, periods 1–4, C7-215

Lecturer: Nguyen Tuan Dung

Group members: Luu Hieu An - 202400093
Nguyen Phu An - 20235466
Pham Chi Bang - 20235477
Nguyen Thanh Lam - 20235518

Hanoi, 2026

Contents

Abstract

Video people counting is an important computer vision problem for crowd monitoring, traffic analysis, safety management, and real-time camera systems, but applying a CNN detector to every frame is computationally expensive. The problem is also interesting from a parallel-programming perspective because video inference contains many repeated detector calls that can be decomposed across both time and space, while still requiring careful result merging to avoid duplicate counts near tile boundaries. This report presents *MPI-Based YOLO People Counting*, a C++17/OpenMPI pipeline that parallelizes pretrained YOLO inference on a three-MacBook physical cluster. The input video is first split temporally into frames and spatially into image tiles; each frame-tile pair becomes an MPI task assigned by static block-cyclic scheduling. Each rank runs a long-lived local YOLO worker on its assigned tasks, then rank 0 gathers detections, remaps tile-local coordinates to full-frame coordinates, removes duplicates with ownership filtering and global non-maximum suppression, computes frame-level people counts, and writes CSV timing artifacts. Dynamic master-worker scheduling is evaluated only as a comparison because its extra task-dispatch communication performs worse for this workload. The experiments separate parallel correctness from detector accuracy: the MPI implementation matches the serial pipeline on 30 tested frames with zero mismatches, while YOLO undercounts crowded MOT17-derived frames with a mean absolute count error of 7.983 and an average prediction of 8.223 people versus a ground-truth average of 16.207. For the required workload scale, $N = 600$ frames is selected because the 12-process run takes 123.667 seconds, and the $2N = 1200$-frame weighted 4/6/2 placement reaches 129.852 seconds with 2.662x wall-clock speedup. Overall, the results show that video people counting can be parallelized effectively as a task-level MPI workload, but performance depends strongly on communication overhead, tile granularity, and rank placement across heterogeneous machines.

Chapter 1

Introduction

Video people counting is a practical computer vision problem with applications in crowd monitoring, traffic analysis, safety management, and camera-based automation. A modern detector such as YOLO can identify people in individual images, but applying it repeatedly to every frame of a video quickly becomes computationally expensive. This makes the problem useful for a parallel-programming study: it is realistic enough to require a full inference pipeline, yet structured enough to expose clear opportunities for decomposition, scheduling, communication analysis, and speedup measurement.

This project focuses on inference rather than model training. The detector itself is a pretrained YOLO model, while the parallel-programming work is the C++17/OpenMPI runtime built around it. The system decomposes a video temporally into frames and spatially into image tiles, then treats each frame-tile pair as an MPI task. Each rank runs YOLO on its assigned tasks through a long-lived local worker process. After inference, rank 0 gathers detections, converts tile-local bounding boxes back to full-frame coordinates, removes duplicate detections near tile boundaries, computes frame-level people counts, and writes the experiment artifacts.

The report evaluates the implementation as a distributed-memory parallel program and separates parallel correctness from detector accuracy. Parallel correctness asks whether the MPI implementation preserves the output of the serial version of the same pipeline, while detector accuracy asks whether YOLO's predicted counts match human ground truth from MOT17-derived data. This distinction is important because the detector may undercount people in crowded scenes even when the MPI program is correct. The experiments therefore measure correctness, input-size suitability, scheduling behavior, granularity, speedup, and heterogeneous rank placement.

The main contributions and insights of this technical report are:

1. It formulates YOLO video people counting as a task-parallel MPI workload over frames and image tiles.
2. It presents an OpenMPI implementation that combines static block-cyclic task mapping, rank-local YOLO workers, centralized detection merging, duplicate removal, and metric collection.
3. It verifies that the MPI pipeline preserves serial output in the correctness experiment, while also showing that detector accuracy against ground truth is a separate limitation.
4. It compares scheduling and placement choices, showing that static scheduling and weighted rank placement are more suitable than dynamic dispatch for the measured heterogeneous cluster workload.
5. It analyzes speedup and remaining bottlenecks, especially communication overhead, tile granularity, master-side post-processing, and unequal machine capacity.

Chapter 2

Problem Description

2.1 CNN-Based Computer Vision Inference

The problem is to execute CNN-based video inference faster by distributing repeated detector work across MPI processes. The program applies a fixed pretrained model; it does not train or update model weights. The parallel system must preserve the output of the serial inference pipeline while reducing runtime or revealing the cost of communication.

2.2 YOLO Video People Counting

The input video is a finite sequence of frames:

$$V = \{F_1, F_2, \ldots, F_N\}. \tag{2.1}$$

The target output for each frame is the number of detected people after post-processing:

$$count(F_i) = |\{b \mid b \text{ is a final person bounding box in } F_i\}|. \tag{2.2}$$

The detector backend is a pretrained YOLO11n model through a local Python worker process [4]. YOLO11 follows the usual one-stage detector structure with a feature-extraction backbone, a feature-fusion neck, and multi-scale detection heads, as shown in Figure 2.1. The MPI program does not split YOLO neural-network kernels. Instead, MPI distributes repeated YOLO calls over frame-tile tasks.

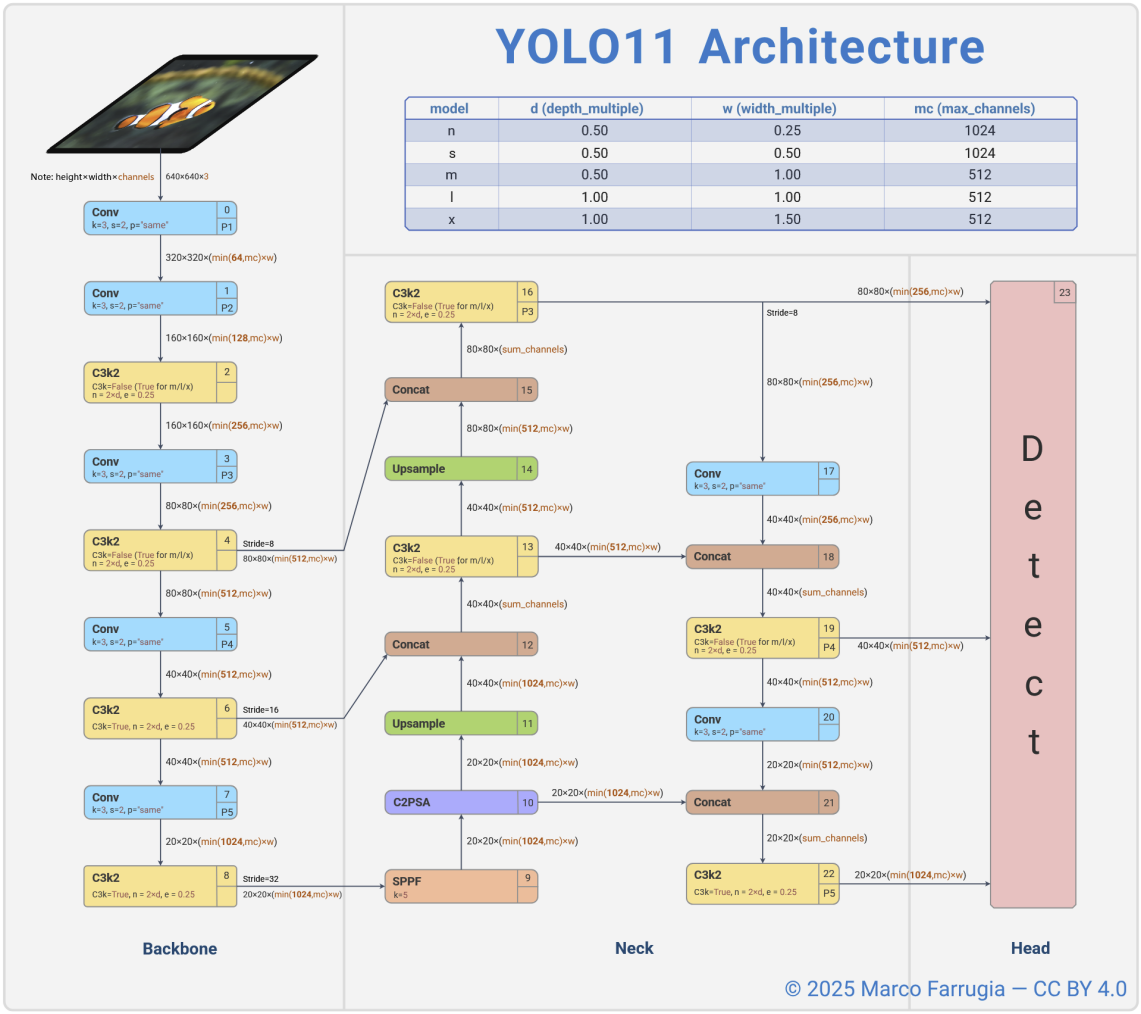


Figure 2.1: YOLO11 object-detection architecture used as the detector backend in this project. Source: Farrugia, CC BY 4.0, Zenodo DOI 10.5281/zenodo.17508018 [1].

For a tile grid with R rows and C columns, frame F_i is decomposed into:

$$F_i \rightarrow \{T_{i,1}, T_{i,2}, \ldots, T_{i,R\times C}\}. \tag{2.3}$$

A task is one frame-tile inference:

$$task(i,j) = T_{i,j}, \qquad |\mathcal{T}| = N \times R \times C. \tag{2.4}$$

Rank 0 converts tile-local detections back to full-frame coordinates, applies tile ownership filtering, removes duplicates, and writes final frame counts.

2.3 Parallel-Programming Focus

Table 2.1: Parallel-programming concepts applied in the YOLO MPI pipeline

Concept	Application in the YOLO MPI implementation
Parallelism level	Task-level parallelism. Each independent task corresponds to running YOLO inference on one frame tile.
Decomposition technique	Hybrid data decomposition: temporal decomposition over video frames combined with spatial decomposition over image tiles. The original video is decomposed into a one-dimensional list of frame-tile tasks.
Task granularity	Tile-level granularity. One task contains the inference workload for one tile of one frame. The tile size and number of tiles control whether the workload is fine-grained or coarse-grained.
Mapping technique	One-dimensional processor assignment using a flattened task list. Tasks are assigned to MPI ranks by static block-cyclic mapping. In the weighted version, faster devices receive a larger share of tasks.
Communication strategy	Communication is minimized during inference. Each rank performs local computation independently, then sends its local detections and timing metrics to rank 0 for global merging.
Communication topology	Master–worker topology. Worker ranks compute local YOLO detections, while rank 0 acts as the master process for gathering results, applying global post-processing, and writing output files.
Communication mode	Blocking collective communication is used in the current implementation during the gather phase. There is no communication between ranks during local YOLO inference.
Load balancing considerations	Load balancing is handled through tile-level task granularity, block-cyclic task distribution, and weighted process placement. The goal is to reduce idle time between MPI ranks, especially when machines have different computing speeds.
Correctness verification	The parallel result is considered correct if the final frame-level people counts match the serial YOLO pipeline after applying the same confidence filtering, coordinate remapping, and non-maximum suppression.
Performance evaluation	Execution time is measured with and without communication time. Speedup is evaluated by increasing the number of MPI processes while keeping the input size fixed at the selected benchmark size.

2.4 Correctness Targets

The serial YOLO baseline processes the same frame-tile task list without MPI distribution and applies the same post-processing pipeline. Parallel correctness is:

$$\Delta_i = |count_{parallel}(F_i) - count_{serial}(F_i)|. \tag{2.5}$$

The correctness experiment reports mismatched frames, maximum count error, and mean count error. Detector accuracy against MOT17 ground truth is evaluated separately because model error and parallelization error are different questions.

Chapter 3

Parallelization Methods

3.1 Task Parallel YOLO Inference

The YOLO pipeline uses task-level parallelism over independent frame-tile inference tasks. The implementation is centered on `src/yolo_mpi_cpp.cpp` in the repository. Each MPI rank runs the same C++ program, and rank-local detector work is delegated to a long-lived Python YOLO worker.

3.1.1 Temporal-Spatial Decomposition

The input video is first decomposed by time into frames and then by space into image tiles. If the video has N frames and each frame is split into an $R \times C$ grid, the total number of tasks is NRC. Figure 3.1 compares the original frame with 2x2 and 5x4 tiling. The 2x2 grid keeps detector accuracy close to the original-frame baseline because it introduces fewer artificial tile borders, so fewer people are cut by tile edges before the final merge stage. Larger grids create more tasks and can improve scheduling flexibility, but they increase duplicate detections near tile borders and increase post-processing work.

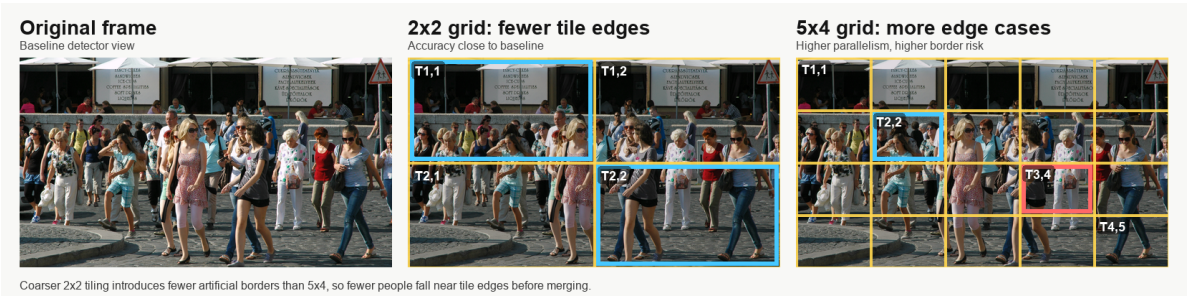


Figure 3.1: Hybrid temporal-spatial decomposition for YOLO video inference. The public-domain crowd frame is from Wikimedia Commons [2]; the overlays show why coarse 2x2 tiling preserves accuracy better than finer 5x4 tiling while still creating independent frame-tile tasks.

3.1.2 1D Mapping and Static Scheduling

All frame-tile tasks are flattened into a one-dimensional list. For task index i, chunk size k, and P ranks, static block-cyclic scheduling assigns:

$$owner(task_i) = \left\lfloor \frac{i}{k} \right\rfloor \bmod P. \tag{3.1}$$

Static scheduling has low dispatch overhead because ranks can determine their own tasks locally. The measured repository results show that this advantage matters for the 600-frame 5x4 tiled YOLO workload.

3.1.3 Communication and Merging

After each rank finishes its assigned frame-tile tasks, it sends two types of data to rank 0: detected bounding boxes and timing metrics. The communication pattern is centralized. Worker ranks compute locally, then rank 0 gathers the local result payloads and performs the final merge.

For each detected person box, the worker stores:

$$(frame\_id,\ tile\_id,\ x_1,\ y_1,\ x_2,\ y_2,\ score,\ class). \tag{3.2}$$

The box coordinates are still relative to the local tile. Rank 0 first converts them back to original-frame coordinates:

$$x^{global} = x^{local} + x_s, \qquad y^{global} = y^{local} + y_s, \tag{3.3}$$

where (x_s, y_s) is the top-left coordinate of the tile inside the original frame.

For each frame F_i, rank 0 merges all detections from the $R \times C$ tiles:

$$B_i^{raw} = \bigcup_{j=1}^{RC} B_{i,j}. \tag{3.4}$$

Since a person near a tile boundary may appear in more than one tile, duplicate boxes can occur. Rank 0 therefore applies tile-owner filtering followed by global non-maximum suppression:

$$B_i^{final} = NMS(B_i^{raw}, \theta). \tag{3.5}$$

The final people count for frame F_i is:

$$count(F_i) = |B_i^{final}|. \tag{3.6}$$

This centralized merge is necessary because each rank only sees its own assigned tiles, while duplicate removal requires all detections from the same frame.

3.1.4 YOLO Algorithm Sketch

Algorithm 1 Static MPI YOLO people-counting pipeline

Input video contains N frames; each frame is split into an $R \times C$ grid.
Build task list $\mathcal{T}$, where each task is $(frame\_id, tile\_id)$.
Flatten $\mathcal{T}$ into a 1D list and assign task index q using

$$owner(q) = \left\lfloor \frac{q}{K} \right\rfloor \bmod P,$$

where K is chunk size and P is the number of MPI ranks.
for each MPI rank r in parallel **do**
  Select all tasks whose owner is r.
  for each selected task (i, j) **do**
    Crop tile j from frame F_i.
    Run YOLO inference on the tile.
    Keep only `person` detections.
    Store local boxes and timing metrics.
  end for
end for
Gather serialized detections and rank metrics to rank 0.
if rank is 0 **then**
  for each frame F_i **do**
    Convert tile-local boxes to global frame coordinates.
    Merge boxes from all tiles of frame F_i.
    Apply tile-owner filtering and global NMS.
    Compute $count(F_i) = |B_i^{final}|$.
  end for
  Write per-frame counts and runtime metrics to CSV files.
end if

3.2 Weighted Process Placement

The cluster is heterogeneous because the three machines do not have the same CPU throughput. If every machine receives the same number of MPI ranks, the faster machines may finish early and wait for the slower machine. This creates idle time and increases the total runtime.

Weighted process placement keeps the same static block-cyclic task mapping, but changes how many MPI ranks are launched on each physical machine. A stronger machine receives more ranks, so it receives more positions in the global task stream. A weaker machine receives fewer ranks, so it receives less work. The report compares uniform 4/4/4 placement against weighted placements such as 3/6/3 and 4/6/2. Here, 4/6/2 means the master runs 4 MPI ranks, node1 runs 6 MPI ranks, and node2 runs 2 MPI ranks.

The scheduling formula does not change:

$$owner(q) = \left\lfloor \frac{q}{K} \right\rfloor \bmod P. \tag{3.7}$$

Only the mapping from MPI ranks to physical machines changes. Therefore, if runtime and idle gap improve, the improvement comes from better process placement, not from a different scheduling algorithm.

3.3 Metric Collection

Each experiment records both correctness and timing metrics. Correctness is checked using the final people counts for each frame. Timing metrics are collected to explain where the runtime is spent.

The main recorded metrics are:

$$T_{wall}, \tag{3.8}$$

the total end-to-end runtime;

$$T_{compute,r}, \tag{3.9}$$

the YOLO inference and local processing time of rank r;

$$T_{comm,r}, \tag{3.10}$$

the time rank r spends sending or receiving results; and

$$T_{no\_comm} = T_{wall} - T_{comm}. \tag{3.11}$$

These metrics are necessary because speedup alone is not enough. A run can have many parallel tasks but still be slow if rank 0 spends too much time gathering detections, or if a slow machine receives too many tasks.

3.4 Load Balancing and Metrics

For each rank r, the total rank time is modeled as:

$$T_r = T_{compute,r} + T_{comm,r} + T_{idle,r}. \tag{3.12}$$

The idle-gap indicator is:

$$idle\_gap = \frac{T_{max} - T_{min}}{T_{max}}, \tag{3.13}$$

where $T_{max} = \max_r T_r$ and $T_{min} = \min_r T_r$. A smaller idle gap means the workload is distributed more evenly. The course criterion treats a run as balanced when:

$$idle\_gap \leq 0.25. \tag{3.14}$$

In the YOLO experiments, equal process placement such as 4/4/4 does not satisfy this condition, while the best weighted static placement reduces idle time and satisfies the load-balancing criterion.

Chapter 4

Experimental Setup

4.1 Repository Evidence

The report is based on the local repository `yolo-mpi-people-count`. The strongest source for measured YOLO values is `reports/final_report_hust_style.md`, with additional full-cluster discussion in `reports/additional_experiments_20260623.md`. The artifact branch also provides compact CSV files under `reports/artifacts/`; the YOLO MPI CSV artifacts have been copied into this report repository under `artifacts/`.

4.2 Physical Cluster

The project uses three physical MacBook machines connected through a local network. The benchmark mode uses CPU execution to match the course emphasis on MPI processes and processor assignment. GPU/MPS execution is kept for demonstration and live-camera use rather than formal CPU speedup measurement.

Table 4.1: Physical cluster roles

Host	Machine type	Role in experiments
master	MacBook Air M4	Coordinates execution, gathers results, stores outputs, and can run local tasks.
node1	MacBook Pro M4	Worker host; measured results suggest it can sustain more assigned ranks.
node2	MacBook Air M2	Worker host; weighted placement assigns fewer ranks to reduce imbalance.

4.3 Software and Data

The implementation uses C++17, OpenMPI, Python helper scripts, and a pretrained Ultralytics YOLO detector. The main dataset source is MOT17-derived video data [3]. Compact MOT17 subsets are used for quick correctness and granularity experiments; longer sequence segments are used for the required input-size and speedup runs.

Table 4.2: Main YOLO experimental configuration

Item	Setting
Detector	Pretrained YOLO11n, person-class filtering
Parallel framework	C++17 and OpenMPI
Benchmark device	CPU
Dataset	MOT17-derived sequences
Main process count	12 MPI processes
Main long-run grid	5x4 regions
Scheduler	Static block-cyclic task assignment
Placement variants	Uniform 4/4/4 and weighted 3/6/3, 4/6/2

4.4 Metrics

The report uses the following metrics from the project reports and generated CSV design:

- serial-versus-MPI correctness: mismatched frames and count error;
- detector accuracy: MAE, RMSE, percentage error, exact-match ratio;
- runtime with communication: wall-clock time including coordination;
- runtime without communication: compute-oriented time reported by the scripts;
- speedup and efficiency:

$$S_p = \frac{T_1}{T_p}, \qquad E_p = \frac{S_p}{p}; \tag{4.1}$$

- idle-gap indicator for load-balance analysis.

Chapter 5

Results and Discussion

5.1 Detector Accuracy Under Tile Splitting

This experiment checks whether splitting each frame into spatial tiles changes YOLO detector accuracy against MOT17 ground-truth counts. It evaluates detector behavior under tiling, not parallel speedup or latency.

Table 5.1: YOLO accuracy comparison across baseline and tile-splitting configurations

Configuration	Grid	Frames	Runtime (s)	MAE	RMSE	Percent err.	Pred. avg.
Single-machine baseline	1x1	300	–	7.983	8.269	0.490	8.223
MPI tiled	2x2	300	–	7.983	8.269	0.490	8.223
MPI tiled	4x3	300	–	8.223	8.517	0.504	7.977
MPI tiled	5x4	300	54.764	8.383	8.682	0.514	7.812

The runtime entry is included only where the repository contains an official measurement for the same 300-frame workload, so this table should not be used to claim latency improvement for 2x2 tiling. The 2x2 configuration has the same MAE, RMSE, percent error, and predicted average count as the single-machine baseline, meaning moderate tiling preserves detector behavior on this subset. Finer grids slightly degrade accuracy: MAE rises to 8.223 for 4x3 and 8.383 for 5x4, while the predicted average count falls from 8.223 to 7.812. This suggests that aggressive tiling introduces border effects, reduced image context, and imperfect duplicate handling near tile boundaries.

5.2 Input-Size Selection

This experiment chooses the main workload size required by the course. Here, “Frames” means the number of video frames processed in one run, not the number of videos. The table should be read as input-size selection rather than a perfectly controlled scaling experiment, because the compact quick tests use a 4x3 grid while the longer runs use the final 5x4 grid.

Table 5.2: YOLO input-size selection

Frames	P	Grid	Tasks	Runtime with comm. (s)	Runtime without comm. (s)	ms/task	Selection note
30	12	4x3	360	9.165	2.999	25.46	Compact quick test
60	12	4x3	720	12.570	6.801	17.46	Compact quick test
100	12	4x3	1200	17.776	11.596	14.81	Compact quick test
150	12	4x3	1800	23.585	17.103	13.10	Compact quick test
300	12	5x4	6000	54.764	49.156	9.13	Long sequence
600	12	5x4	12000	123.667	114.352	8.97	Selected N
837	12	5x4	16740	146.316	139.136	8.74	Best ms/task, longer validation
1200	12	5x4	24000	129.852	128.122	5.41	$2N$, weighted run

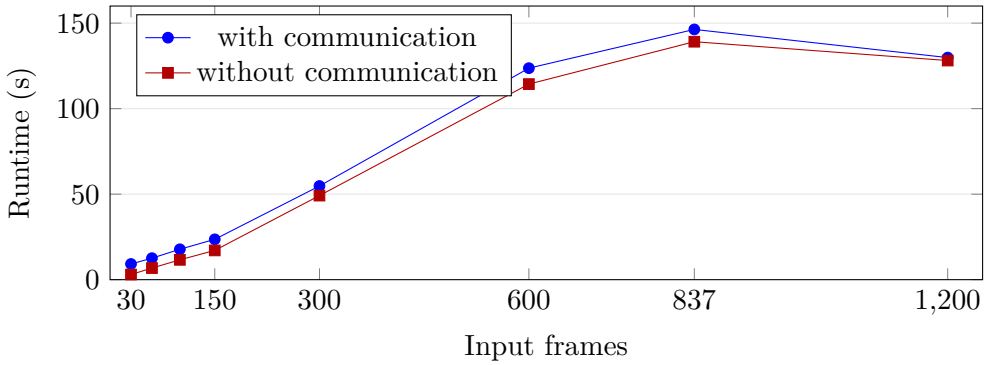


Figure 5.1: Runtime versus input size. The 600-frame workload is the first measured case inside the required 120–180 second interval.

The time-per-task column shows that very small inputs have high overhead because startup, synchronization, and aggregation are amortized over too few tasks. The selected workload is $N = 600$ frames because it is the first long-sequence case whose 12-process wall-clock runtime, 123.667 seconds, satisfies the required 120–180 second range. The 1200-frame row is used as the $2N$ speedup workload and uses weighted placement, so it should not be compared naively against the 600-frame row as if only the frame count changed.

5.3 Granularity and Load Balance

This experiment asks whether finer tile granularity creates a more balanced workload and how much communication it adds. The weighted placement keeps the idle-gap indicator at 0.235, within the 25 percent course criterion, but increasing the number of regions still increases task count and communication.

Table 5.3: YOLO granularity and load balance with weighted 6/4/2 placement

Grid	P	Tasks	Task/rank	Max comp.	Comm.	Comm/task	Idle gap	Balance
1x1	12	150	12.5	0.731	10.674	71.16	0.235	Balanced
2x2	12	600	50.0	3.155	16.306	27.18	0.235	Balanced
4x3	12	1800	150.0	9.266	34.881	19.38	0.235	Balanced
5x4	12	3000	250.0	13.733	51.790	17.26	0.235	Balanced

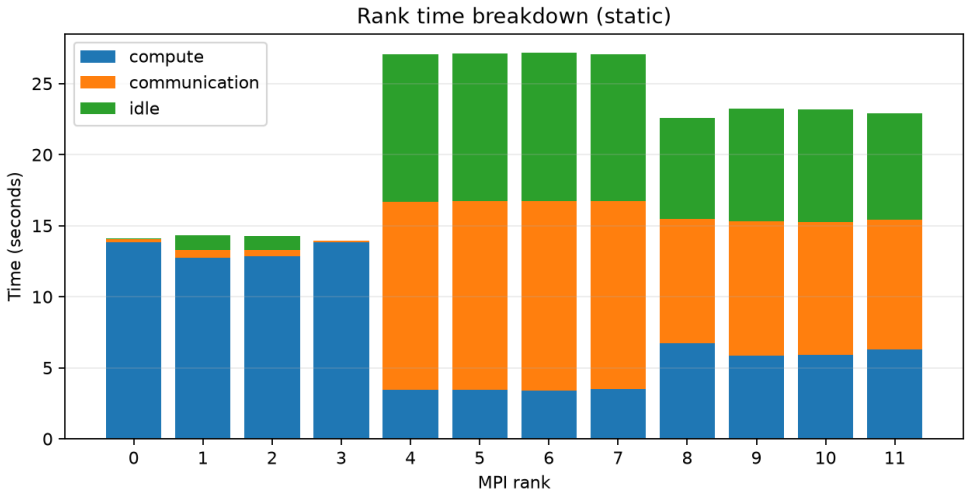


Figure 5.2: Per-process timing breakdown for the granularity experiment. Each bar is one MPI process, with compute, communication, and idle time stacked in the same column.

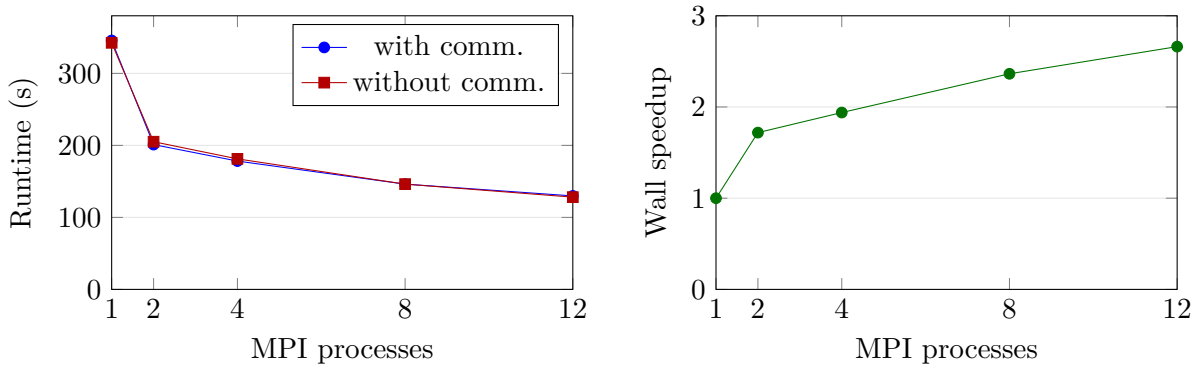
Increasing the tile grid increases the number of frame-tile tasks from 150 at 1x1 to 3000 at 5x4. This gives the scheduler more work units, but total communication also rises from 10.674 seconds to 51.790 seconds and the master must merge more intermediate detections. The idle-gap indicator remains 0.235 under the weighted placement, so the result supports two claims: weighted placement can keep the workload balanced, but finer granularity is not automatically faster because communication and post-processing overhead also increase. Figure 5.2 shows why the load-balance check must inspect process-level timing rather than only total runtime.

5.4 Speedup and Efficiency

The speedup experiment uses the selected input size doubled to $2N = 1200$ frames and applies weighted static placement for the multi-rank runs. This keeps the process distribution consistent with the heterogeneous cluster: stronger machines receive more ranks, while the slower machine receives fewer ranks.

Table 5.4: YOLO speedup on 1200 frames with weighted placement

P	Runtime with comm. (s)	Runtime without comm. (s)	Wall speedup	Efficiency	Note
1	345.677	342.467	1.000	1.000	Baseline
2	201.111	205.100	1.719	0.859	Weighted placement
4	178.238	181.289	1.939	0.485	Weighted placement
8	146.212	146.204	2.364	0.296	Weighted placement
12	129.852	128.122	2.662	0.222	Weighted 4/6/2



(a) Runtime by process count

(b) Wall-clock speedup

Figure 5.3: Runtime and speedup trends for the 1200-frame weighted-placement run.

The speedup is measurable but sublinear. Wall-clock speedup rises from 1.719x at two processes to 2.662x at twelve processes, showing that the MPI pipeline benefits from distributing frame-tile inference across machines. However, efficiency drops as P increases, so each added process contributes less marginal benefit because of communication, aggregation at rank 0, heterogeneous node speeds, and duplicated work around tile boundaries.

5.5 Weighted Static Placement

This experiment tests whether equal rank placement is appropriate on the heterogeneous three-MacBook cluster. The best measured weighted placement assigns four ranks to the master, six to node1, and two to node2, giving more ranks to the stronger machine and fewer ranks to the slower one.

Table 5.5: Weighted static placement on heterogeneous cluster

Placement	Distribution	Runtime with comm. (s)	Runtime without comm. (s)	Idle gap	Balance
Uniform	4/4/4	40.619	36.522	0.466	Not balanced
Weighted	3/6/3	34.712	30.339	0.371	Not balanced
Weighted	4/6/2	29.581	27.484	0.235	Balanced

Weighted placement is the strongest load-balancing result in the report. Uniform 4/4/4 placement has a higher idle gap and fails the 0.25 balance criterion. Moving to weighted placements improves runtime and idle gap because stronger machines receive more MPI ranks while the slower machine receives fewer ranks. The best measured placement is 4/6/2, which reduces the idle gap below the balance threshold. This supports the claim that processor assignment matters: equal rank counts are not necessarily fair when hosts have different throughput.

5.6 YOLO Parallel Correctness

This experiment verifies parallel correctness: after task assignment, MPI gathering, coordinate remapping, duplicate removal, and final counting, the MPI output should match the serial

pipeline output.

Table 5.6: YOLO serial-versus-MPI correctness

Result	Schedule	Grid	P	Frames	Mismatch	Max err.	Mean err.	Evidence
Passed	Static	4x3	12	30	0	0	0.000	correctness CSV

The MPI run has zero mismatched frames over the 30-frame verification sequence. Both the maximum count error and mean count error are zero, so the parallel pipeline exactly reproduces the serial pipeline for this tested configuration. This proves serial-versus-MPI consistency; it does not prove that YOLO is accurate against human ground truth, which is measured separately in the detector accuracy section.

5.7 Summary of Findings

The measured YOLO results support the main task-parallel narrative. The implementation preserves serial output, reaches the required input-size runtime, shows measurable speedup, and demonstrates that task granularity and rank placement strongly affect performance. The strongest measured improvement in load balance comes from weighted static placement, which reduces the idle-gap indicator from 0.466 to 0.235.

Chapter 6

Conclusion

This report presented a task-parallel YOLO-based people-counting pipeline using MPI. The method decomposes the input video both temporally and spatially: the video is divided into frames, and each frame is further divided into image tiles. Each frame-tile pair becomes an independent inference task. These tasks are assigned to MPI ranks using static block-cyclic mapping, processed locally by YOLO workers, and finally merged at rank 0 using coordinate remapping, tile-owner filtering, and global non-maximum suppression.

The main strength of the implementation is its simplicity and clear parallel structure. The workload is naturally parallel because YOLO inference on different frame tiles can be executed independently. The correctness test shows that the MPI version produces the same frame counts as the serial tiled version, with zero mismatched frames. The selected 600-frame workload runs in 123.667 seconds using 12 processes, which satisfies the required runtime range. In the 1200-frame experiment, the weighted 4/6/2 placement achieves 129.852 seconds and a 2.662x wall-clock speedup. It also reduces the idle-gap indicator to 0.235, satisfying the 25 percent load-balancing criterion.

However, the method also has limitations. First, the centralized merging stage at rank 0 can become a communication and post-processing bottleneck when the number of detections increases. Second, static scheduling has low overhead but cannot fully adapt to uneven machine speeds or variable inference time across frames. Third, tiling may create duplicate detections near tile boundaries and can reduce detection quality because each tile contains less visual context than the full frame. The recorded accuracy experiment shows that YOLO undercounts crowded MOT17 frames, with MAE 7.983. This is mainly a detector limitation, not an MPI correctness error.

Future work can improve the method in three directions. First, dynamic or hybrid scheduling could be used to assign more work to faster ranks during runtime. Second, hierarchical merging could reduce the bottleneck at rank 0 by aggregating detections at node-level leaders before sending results to the master rank. Third, the detection quality could be improved by using overlap-aware tiling, better duplicate-removal rules, or a stronger detector for dense crowd scenes.

Overall, the project demonstrates the key parallel programming concepts required by the course: decomposition, task mapping, communication strategy, scheduling, load balancing, correctness verification, and speedup measurement. The final implementation is not perfect, but it provides a coherent and measurable MPI parallelization method for CNN-based computer vision inference.

Bibliography

[1] Marco Farrugia. Yolo11 architecture diagram: Visual representation of the yolo11 object detection model. https://doi.org/10.5281/zenodo.17508018, 2025.

[2] Keymeulen. Crowd crossing the street, hungary 2011. https://commons.wikimedia.org/wiki/File:Crowd_crossing_the_street,_Hungary_2011.jpg, 2011. CC0 1.0 Universal Public Domain Dedication.

[3] MOTChallenge. Mot17 benchmark. https://motchallenge.net/data/MOT17/, 2026.

[4] Ultralytics. Ultralytics yolo documentation. https://docs.ultralytics.com/, 2026.